

Optimization III

Inequality Constraints

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Lagrangian method and Kuhn-Tucker conditions

- FOC
- SOC
- Multiple constraints, nonnegativity constraints
- Some research applications

2 variables - 1 constraint

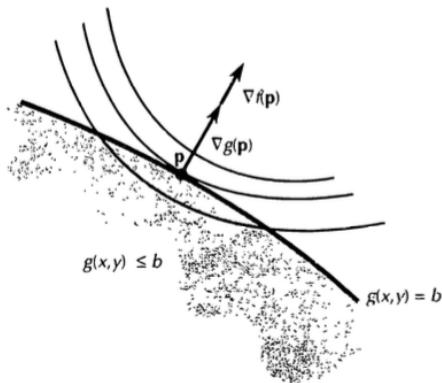
Canonical form

$$\begin{aligned} & \max f(x, y) \\ & s.t. \ g(x, y) \leq b \end{aligned}$$

or

$$\begin{aligned} & \min f(x, y) \\ & s.t. \ h(x, y) \geq b \end{aligned}$$

Consider the max problem.

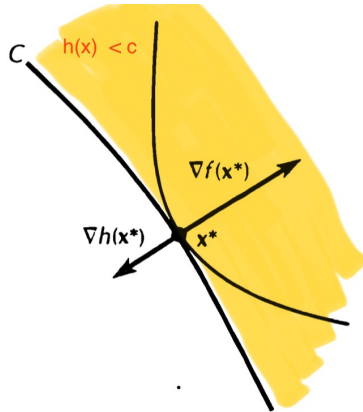
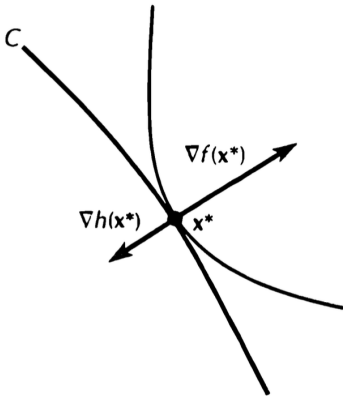


∇f and ∇g point in the same direction at the maximizer p .

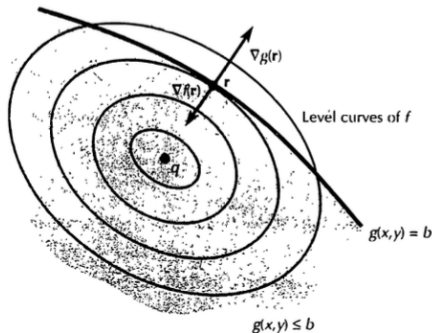
- If the optimal point is p lying on $g(x, y) = b$, the constraint is binding/active.
- $\nabla f(p) - \lambda \nabla g(p) = 0$. The sign of λ must be positive, i.e., the two gradient vectors point in the same direction.

What happens if the two vectors point in opposite directions?

Consider $\max f(x, y)$ subject to $h(x, y) \leq c$.



The critical point (x^* in the figure) is no longer a max. In this case, no max point can be determined because we can increase f as much as we want while the



The situation in which the constraint is not binding.

- If the optimal point is q lying inside the region $g(x, y) < b$, the constraint is not binding/inactive.
- When the constraint is inactive, we can treat q as the optimal point in an unconstrained problem, and $\nabla f(q) = 0$. At the same time, setting $\lambda = 0$ removes the constraint from the Lagrangian.

- If the constraint is binding, then $g(x, y) - b = 0$ and $\lambda > 0$
- If the constraint is not binding, then $g(x, y) - b < 0$ and $\lambda = 0$.

These two conditions can be written compactly as the **complementary slackness condition**:

$$\lambda \cdot (g(x, y) - b) = 0$$

This condition has a complementary nature, referring to the fact that at most one of the two factors equals 0.

Kuhn-Tucker Formulas

Put the problem in standard form $\max f(x, y)$ s.t. $g(x, y) \leq b$.

Lagrangian function

$$\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda(g(x, y) - b)$$

FOC:

- ① $\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial y} = 0$.
- ② $\lambda(b - g(x, y)) = 0$.
- ③ Complementary slackness: $\lambda \geq 0$ and $g(x, y) \leq b$.

The third condition can be stated as follows:

- When $g(x, y) = b$, then $\lambda \geq 0$.
- When $g(x, y) < b$, then $\lambda = 0$.

Example 1



Problem

$$\begin{aligned} \max f(x, y) &= xy \\ \text{s.t. } g(x, y) &= x^2 + y^2 \leq 1. \end{aligned}$$

Lagrangian function

$$L(x, y, \lambda) = xy - \lambda(x^2 + y^2 - 1)$$

KKT FOC

$$\begin{aligned} (x) : y - 2\lambda x &= 0, \\ (y) : x - 2\lambda y &= 0, \\ (\lambda) : \lambda(x^2 + y^2 - 1) &= 0, \\ (CS) : x^2 + y^2 &\leq 1, \lambda \geq 0. \end{aligned}$$

Solve the two cases: binding and non-binding constraint.

- If the constraint is not binding, $\lambda = 0$, the first two equations imply $x = y = 0$. This solution satisfies all KKT conditions.
- If the constraint is binding, $\lambda \geq 0$. The first two equations imply

$$\frac{y}{2x} = \frac{x}{2y} = \lambda, \text{ implying } x^2 = y^2$$

The constraint is binding so at the same time

$$x^2 + y^2 = 1$$

Solving: $x = \pm 1/\sqrt{2}, y = \pm 1/\sqrt{2}$. In total we have 4 solutions

$x = 1/\sqrt{2},$	$y = 1/\sqrt{2},$	$\lambda = 1/2,$
$x = -1/\sqrt{2},$	$y = -1/\sqrt{2},$	$\lambda = 1/2,$
$x = 1/\sqrt{2},$	$y = -1/\sqrt{2},$	$\lambda = -1/2,$
$x = -1/\sqrt{2},$	$y = 1/\sqrt{2},$	$\lambda = -1/2$

The last 2 solutions are eliminated because they do not satisfy KKT, so only the first 2 are feasible.

Comparing the 3 solutions $\rightarrow (1/\sqrt{2}, 1/\sqrt{2})$ & $(-1/\sqrt{2}, -1/\sqrt{2})$ are max.

1

$$\begin{aligned} \max f(x, y) &= 4 - \frac{1}{2}x^2 - 4y, \\ \text{s.t. } 6x - 4y &\leq a, a > 0 \end{aligned}$$

2

$$\begin{aligned} \max f(x, y) &= x^2 + y^2 + y - 1, \\ \text{s.t. } x^2 + y^2 &\leq 1. \end{aligned}$$

3

$$\begin{aligned} \max f(x, y) &= \sqrt{x} + \sqrt{y}, \\ \text{s.t. } px + qy &\leq m, \quad p, q, m > 0. \end{aligned}$$

4

$$\begin{aligned} \max f(x, y) &= 2 - (x - 1)^2 - e^{y^2}, \\ \text{s.t. } x^2 + y^2 &\leq 1/2. \end{aligned}$$

There are 3 ways to convert from a max to a min problem.

- 1 **Change the sign of the objective function.** The sign of the constraint and FOC remain the same.
- 2 **Change the sign of the constraint from $\leq$ to $\geq$,** keep the objective function as is, FOC remains the same.
- 3 Keep the objective function and constraint sign as is, FOC remains the same but **$\lambda \leq 0$.**

It is recommended to use the first two approaches.

Note: If following the third condition, the last 2 solutions in [▶▶ Example 1](#) are the 2 solutions for the min problem.

Example 2



Find

$$\begin{aligned}\min f(x, y) &= -x^2 + 2y \\ \text{s.t. } x^2 + y^2 &\leq 1\end{aligned}$$

Rewrite as

$$\begin{aligned}\min f(x, y) &= -x^2 + 2y \\ \text{s.t. } -x^2 - y^2 &\geq -1\end{aligned}$$

Lagrangian

$$L(x, y, \lambda) = 2y - x^2 - \lambda(-x^2 - y^2 + 1)$$

KKT FOC:

$$(i) \frac{\partial L}{\partial x} = 0 \iff -2x + 2\lambda x = 0$$

$$(ii) \frac{\partial L}{\partial y} = 0 \iff 2 + 2\lambda y = 0$$

$$(iii) \lambda \cdot (-x^2 - y^2 + 1) = 0,$$

$$(iv) \lambda \geq 0 \text{ (with } \lambda = 0 \text{ when } x^2 + y^2 < 1)$$

From (i), (ii), we deduce $\lambda = 1, y = -1$. Since $\lambda > 0$, the constraint is binding, we have $x^2 + y^2 = 1$. Since $y = -1$, then $x = 0$. The optimal point is $(0, -1)$.

SOC

- ① If the Lagrangian is concave (convex), then the critical point is a max (min).
 - ▶▶ proof
 - If $f(x, y)$ is a convex function and $g(x, y)$ is linear
- ② We can also use the bordered Hessian but need to distinguish whether the constraint is active (binding) or inactive (non-binding).
 - When the constraint is binding, it is the same as the equality constraint case we studied.
 - For non-binding constraints, the Lagrange multiplier equals 0, so they drop out of the Lagrangian and we ignore them, similar to unconstrained optimization.
 - If all constraints are binding at the critical point, we use the theory from equality-constrained optimization.
 - If all constraints are non-binding at the critical point, use the theory from unconstrained optimization.
 - If only some constraints are binding, $\bar{H}$ will only be bordered by those constraints, and we return to the equality-constrained case.

Apply this knowledge to check SOC for ▶▶ Practice .

Multiple Constraints

Suppose we have an optimization problem with n variables and m inequality constraints

$$\max_{\mathbf{x}} \underbrace{f(x_1, \dots, x_n)}_{\mathbf{x}} \text{ s.t. } g_1(\mathbf{x}) \leq c_1, \dots, g_m(\mathbf{x}) \leq c_m$$

- 1 Write the Lagrangian function

$$\mathcal{L}(\mathbf{x}) = f(\mathbf{x}) - \sum_{j=1}^m \lambda_j (g_j(\mathbf{x}) - c_j)$$

- 2 FOCs

$$\frac{\partial \mathcal{L}}{\partial x_i} = 0 \text{ for each } i = 1, \dots, n.$$

- 3 Complementary slackness for $j = 1, \dots, m$

$$\lambda_j \cdot g_j(\mathbf{x}) = 0.$$

where, if $g_j(\mathbf{x}) = c_j$, $\lambda_j \geq 0$, or if $g_j(\mathbf{x}) < c_j$, then $\lambda_j = 0$.

- 4 Find all points $\mathbf{x} = (x_1, \dots, x_n)$ satisfying FOC and the complementary slackness condition.

Given the problem

$$\begin{aligned} \max \quad & x + 3y - 4e^{-x-y} \\ \text{s.t.} \quad & \begin{cases} 2 - x \geq 2y \\ x - 1 \leq -y \end{cases} \end{aligned}$$

- ❶ Convert the problem to standard form.
- ❷ Write the Lagrangian and FOC for the problem.
- ❸ Find the optimal point. Sol: $(x^*, y^*) = (0, 1)$ with $(\lambda_1, \lambda_2) = (2, \frac{4-e}{e})$.
- ❹ Verify SOC by eliminating all binding constraints and reducing to one variable.

This is a special form of inequality-constrained optimization. Consider the problem

$$\begin{aligned} \max \quad & f(x, y) \\ \text{s.t.} \quad & g(x, y) \leq c, \\ & x \geq 0, \\ & y \geq 0 \end{aligned}$$

Written in standard form:

$$\begin{aligned} \max \quad & f(x, y) \\ \text{s.t.} \quad & g(x, y) \leq c, \\ & -x \leq 0, \\ & -y \leq 0 \end{aligned}$$

Lagrangian

$$L(x, y) = f(x, y) - \lambda[g(x, y) - c] - \mu_1(-x) - \mu_2(-y)$$

KKT

$$(i) \quad L_x = f_x - \lambda g_x + \mu_1 = 0$$

$$(ii) \quad L_y = f_y - \lambda g_y + \mu_2 = 0$$

$$(iii) \quad \lambda(g(x, y) - c) = 0,$$

$$(iv) \quad \mu_1 x = 0,$$

$$(v) \quad \mu_2 y = 0$$

Complementary slackness can be stated as

$$(a) \quad \lambda \geq 0, \text{ with } \lambda = 0 \text{ if } g(x, y) < c$$

$$(b) \quad \mu_1 \geq 0, \text{ with } \mu_1 = 0 \text{ if } x > 0$$

$$(c) \quad \mu_2 \geq 0, \text{ with } \mu_2 = 0 \text{ if } y > 0$$

(1) Under what parameter conditions is $q = 0$? ($a, b, \alpha, \beta > 0$)

$$\max_q \pi(q) = R(q) - C(q)$$

$$s.t. R(q) = aq - bq^2,$$

$$C(q) = \alpha q - \beta q^2,$$

$$q \geq 0$$

(2)

$$\max_{x,y} x + \ln(1 + y)$$

$$s.t. px + y \leq m,$$

$$x \geq 0 \text{ and } y \geq 0.$$

(3)

$$\max_{c,l} \alpha \ln c + (1 - \alpha) \ln(1 - l)$$

$$s.t. c \leq wl + m \text{ and } l \geq 0$$

(1)

$$\begin{aligned} \max \quad & \alpha \ln x + (1 - \alpha) \ln y \\ \text{s.t.} \quad & px + qy \leq m, \\ & x \geq \bar{x} \quad (\bar{x} < m, m, p, q > 0) \end{aligned}$$

(2)

$$\begin{aligned} \max \quad & \alpha x + y - e^x - e^{x+y} \\ \text{s.t.} \quad & x + y \geq 4, \\ & x \geq -1, \\ & y \geq 1 \quad (\alpha > 0) \end{aligned}$$

(3)

$$\begin{aligned} \max \quad & y - x^2 \\ \text{s.t.} \quad & y - x \geq -2, \\ & y^2 \leq x, \\ & y \geq 0 \end{aligned}$$

(4)

$$\begin{aligned} \max \quad & xe^{y-x} - 2ey \\ \text{s.t.} \quad & y \leq 1 + \frac{x}{2}, \\ & x \geq 0, \\ & y \geq 0 \end{aligned}$$

Applications

Voluntary Childlessness Problem

Nowadays, there are many cases of women voluntarily choosing not to have children.

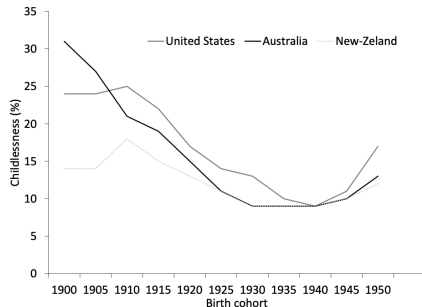
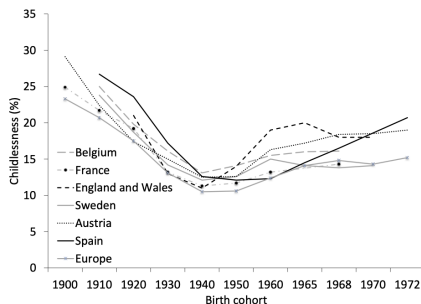


Figure: Childlessness rates in developed countries

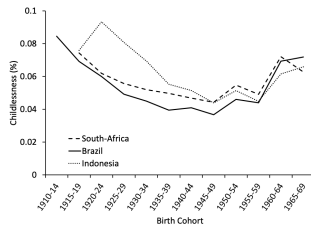
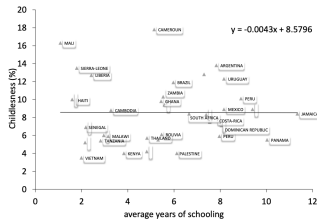


Figure: Childlessness rates in some developing countries

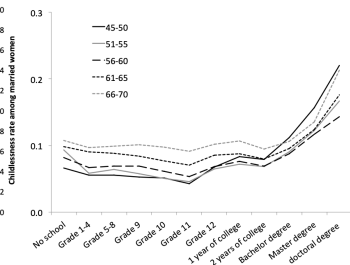
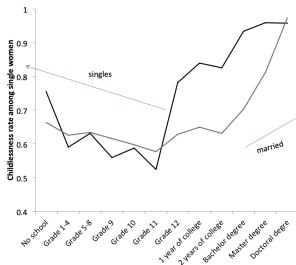


Figure: Childlessness rates and education

We present a simple model to examine this issue. Suppose a woman maximizes the following utility function:

$$\max u(c, n) = \ln c + \delta \ln(n + v)$$

s.t.

$$c \leq (1 - \phi n)w + a,$$

$$n \geq 0$$

- 1 Find (c^*, n^*) to maximize utility.
- 2 When does the woman not want to have children?
- 3 Intuition?

A family decides on consumption c , education spending e , and number of children n to optimize the following utility function

$$\max u(c, n, e) := \ln c + \delta \ln(n \cdot h(e))$$

s.t.

$$c + pen = (1 - \phi n)w,$$

$$h(e) = (e + \theta)^\gamma,$$

$$e \geq 0$$

Given that $p, \phi, \theta, \gamma, \delta > 0$ and $\phi < 1$. Where h is the human capital of the child, θ is public education, and p is education cost. Reduce the problem to 3 variables and 1 constraint.

- ① Find c^*, n^*, e^* to maximize utility for this family. When is $e^* = 0$?
- ② Compare the value of n^* when $e > 0$ and when $e = 0$.
- ③ If base wage w increases, how do decisions about children and education for children change?
- ④ (advanced) Verify optimality using second-order conditions.

A family consists of a wife and a husband. The responsibility of childbearing n and educating children e falls on the woman. The husband devotes all his time to work. The woman's time constraint is

$$l_f + n\phi = 1$$

where l_f is the remaining time the woman spends working. With w_m, w_f being wages for men and women, this family has the utility function:

$$u(c, n, e) = \ln c + \delta \ln n + \delta\gamma \ln(\theta + e)$$

s.t.

$$\begin{aligned} c + en &= w_m + l_f w_f, \\ e &\geq 0. \end{aligned}$$

- 1 Find c^*, n^*, e^* to maximize utility for this family.
- 2 What happens to optimal fertility decisions if the income inequality w_m/w_f narrows?

Appendix

EMEA : Essential mathematics for economic analysis (Syllaeter & Hammond et. al, 6e 2021)

FMMA : Fundamental methods of mathematical economics (Chiang & Wainwright, 4e 2005)

FMEA : Further mathematics for economic analysis (Syllaeter & Hammond et. al, 2e 2008)

ME : Mathematics for economists (Simon & Blume, 1994)

NMO : Numerical methods and optimization (Butenko & Pardalos, 2014)

ORAA : Operations Research: Applications and Algorithms (Winston, 2003)

MA : Microeconomic Analysis (Varian, 1992)

Related further reading: EMEA chap.14.9 (p.569), FMMA chap.3.5 (p.130), ME chap.18.3 p.424 and p.466 (discusses SOC for KKT).

Suppose (x_0, y_0) is a critical point satisfying KKT FOC. If the Lagrangian is concave, then (x_0, y_0) will be a max point. By definition, we can write:

$$\begin{aligned} L(x_0, y_0) &= f(x_0, y_0) - \lambda[g(x_0, y_0) - c] \\ &\geq L(x, y) = f(x, y) - \lambda[g(x, y) - c] \quad \forall (x, y) \end{aligned}$$

This is equivalent to

$$f(x_0, y_0) - f(x, y) \geq \lambda[g(x_0, y_0) - g(x, y)] \quad \forall (x, y) \quad (1)$$

Consider two cases: active and inactive constraints

- If $g(x_0, y_0) < c$, we have $\lambda = 0$, so the right-hand side of (1) equals 0, meaning $f(x_0, y_0) \geq f(x, y)$ for all (x, y) .
- If $g(x_0, y_0) = c$ and $g(x, y) \leq c$, the right-hand side of (1) becomes

$$\lambda[g(x_0, y_0) - g(x, y)] = \lambda[c - g(x, y)] \geq 0$$

because $\lambda \geq 0$ and $c - g(x, y) \geq 0$ for all (x, y) .

- Conclusion: (x_0, y_0) is a max point when it satisfies KKT FOC and the Lagrangian is concave.